



Karnataka State Police
Government of Karnataka

Powered by **H2S**

0atathon 2026

Technology partner



Team Details

- a. Team name: Insight Out
- b. Team leader name: Vanguri Nissy Niveditha
- c. Team size:2
- d. Problem Statement:



Brief about the solution

Imagine a police officer asking, *‘Show me all suspect links in recent Mysuru house burglaries.’* Instead of waiting days for manual cross-referencing, our system instantly executes two things:

1. It routes the request through a **Dual-Engine AI**—fetching exact statistics via **Text-to-SQL** while running semantic **RAG search** across unstructured FIR case files to extract grounded facts with 100% verifiable citations.
2. It dynamically maps a **live criminal network graph** while automatically enforcing strict **Role-Based Access Control (RBAC)**—so a Constable sees only what’s cleared, while an Inspector unlocks the full intelligence web.



Opportunities

- a. How different is it from any of the other existing ideas?
 - b. How will it be able to solve the problem?
 - c. USP of the proposed solution
- **Dual AI Engine:** Intelligently routes prompts to **Text-to-SQL** for exact metrics or **ChromaDB RAG** for deep semantic search across FIR files (BriefFacts).
 - **Visual Networks & Reports:** Instantly generates interactive **node-edge link graphs** for suspect mapping and one-click PDF summaries.
 - **Rank-Aware Security:** Uses **RBAC** (Constable vs. Inspector) to restrict sensitive data and enforces **100% grounded citations** to eliminate hallucinations.



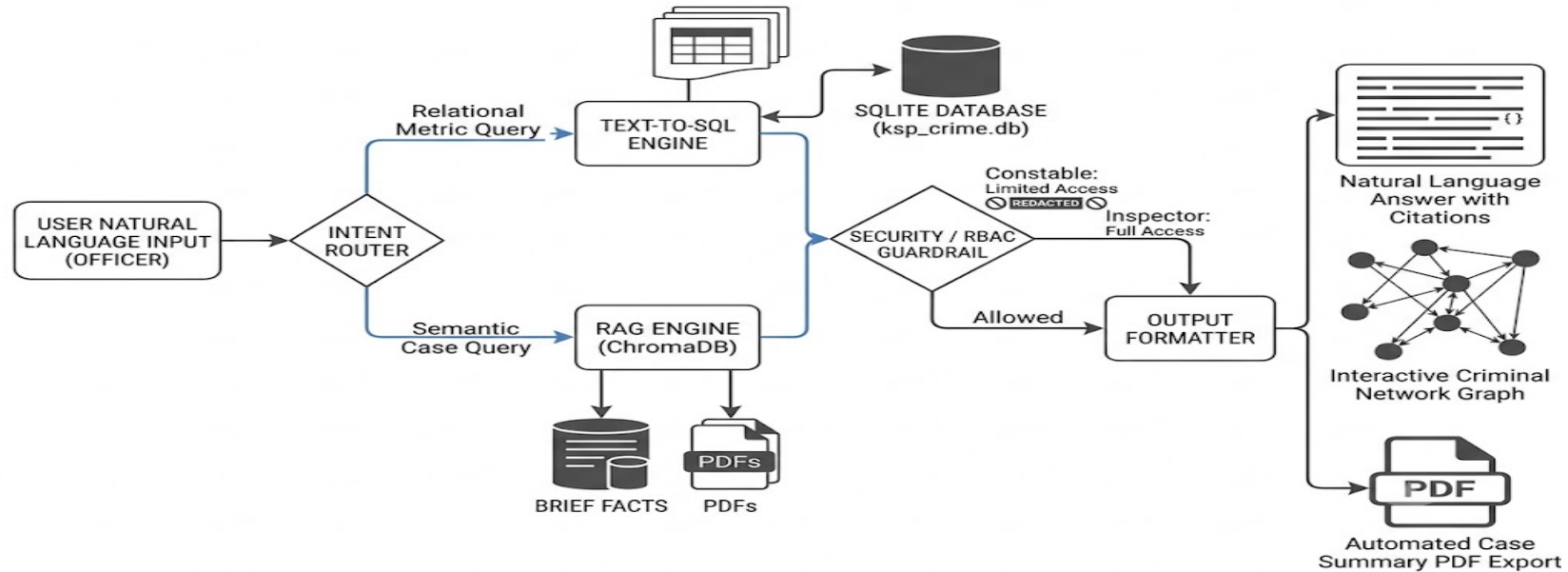
List of features offered by the solution

- Dual-Engine AI Architecture: Combines Text-to-SQL for structured database metrics with Vector RAG (ChromaDB) for deep semantic search across unstructured FIR case notes and BriefFacts.
- Dynamic Visual & Network Analysis: Automatically converts criminal and suspect relationships into interactive node-edge link graphs and provides automated PDF case summary exports.
- Rank-Based Security & Citation Guardrails: Enforces strict Role-Based Access Control (RBAC) to filter sensitive records based on officer rank (Constable vs. Inspector) alongside grounded source citations to prevent hallucinations.



Process flow diagram or Use-case diagram

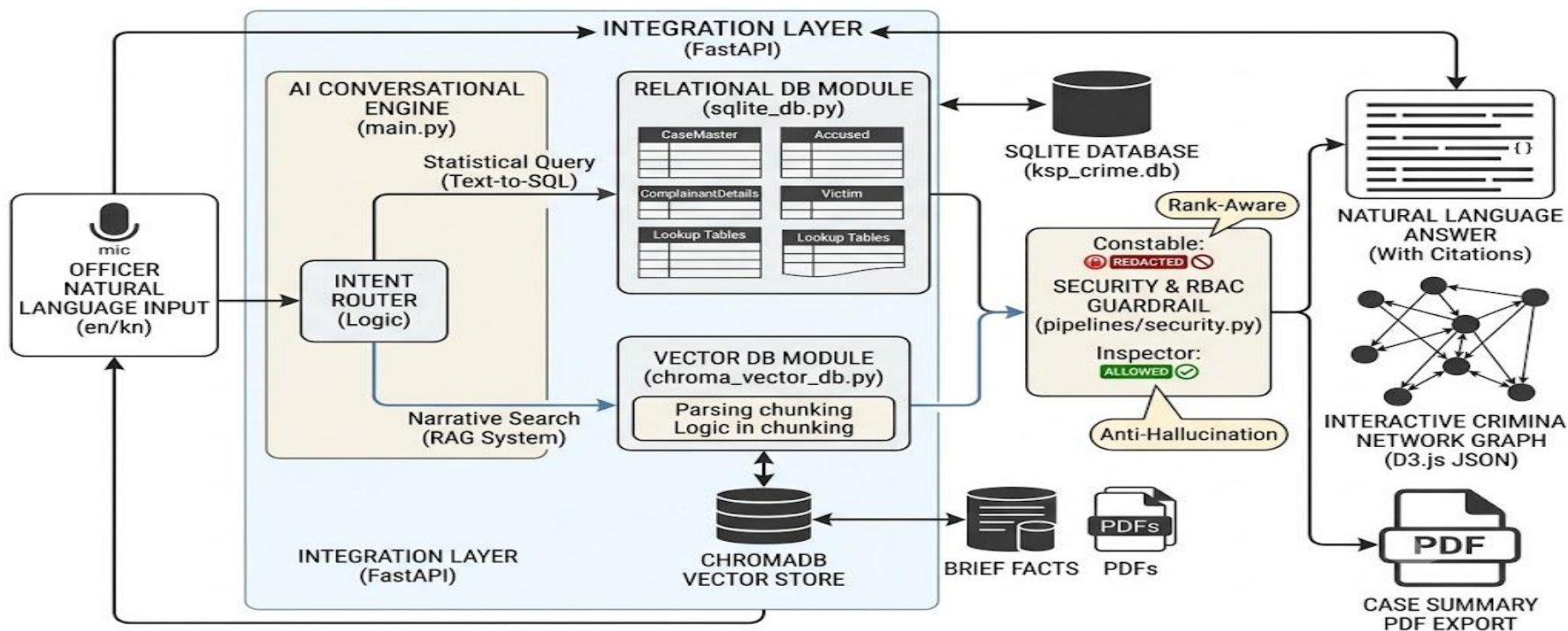
KSP AI CONVERSATIONAL ASSISTANT: COMPLETE BACKEND LOGICAL FLOW





Architecture diagram of the proposed solution

KSP AI CONVERSATIONAL ASSISTANT: COMPLETE SOLUTION ARCHITECTURE





Technologies to be used in the solution

1. Backend & API Orchestration

- Python 3.10+
- FastAPI
- Uvicorn

2. Databases & Vector Storage

- SQLite (ksp_crime.db)
- ChromaDB

3. AI, RAG & Query Engines

- LangChain / LlamaIndex
- LangChain SQL Chain

4. Link Analysis & Visual Outputs

- D3.js / NetworkX / PyVis
- ReportLab / FPDF

5. Security & Access Control

- Custom Security Guardrail Middleware



List down Catalyst Services being used in the solution

- The solution leverages a **Zoho Catalyst** cloud stack to power an intelligent AI conversational assistant for law enforcement.
- **Catalyst Serverless Functions (Python)** and **AppSail** host the backend API and web application, while **QuickML & Zia AI** handle machine learning pipelines, vector embeddings, and RAG search across case files.
- Structured FIR records are managed via the **Catalyst Data Store**, and **SmartBrowz** automates real-time generation of downloadable PDF case summaries—all secured under **Catalyst Authentication** to enforce rank-based access controls (Constable vs. Inspector).



Estimated implementation cost (optional)

Service / Infrastructure Component	Usage Scope	Estimated Cost (INR / Month)
Catalyst Serverless Functions & AppSail	Backend API hosting, routing, & FastAPI execution	₹1,450 – ₹2,900
Catalyst Data Store & Storage	Relational database (ksp_crime.db) + PDF case file storage	₹960 – ₹1,930
QuickML & Zia AI / Vector Search	Embedding generation & ChromaDB vector RAG retrieval	₹2,400 – ₹4,800
SmartBrowz & PDF Generator	Automated rendering & export of case summary PDFs	₹960 – ₹1,450
Catalyst Authentication & RBAC	User session security & rank-based access control rules	₹0 – ₹960
Total Estimated Monthly Cost	Pilot / Station-level Deployment	~₹5,800 – ₹12,000 / month



Snapshots of the prototype

Dashboard

Register FIR

Case History

Reports

Settings

indu

Inspector — Mysuru Central Station

INSPECTOR

Logout

Karnataka Police AI Intelligence System

AI Backend Online Sun, 26 Jul, 2026

indu Inspector

Good evening, indu 🖐️

Inspector · Mysuru Central Station · mysuru

AI Assistant is ready. Query the database in English or Kannada.

KSP AI Engine
SQL · RAG · Graph

TOTAL CASES
1,249

ACTIVE INVESTIGATIONS
313

HIGH-RISK SUSPECTS
48

RESOLVED CASES
888

AI Assistant

Analytics

Crime Map

Network

KSP AI Assistant INSPECTOR

Karnataka State Police Intelligence System

Voice

Export

AI Assistant

Welcome, **indu**. I'm your KSP AI Intelligence Assistant.
You can ask me things like:
"Show all burglary cases in Mysuru"
"How many cases were registered in 2025?"
"Who are the accused in vehicle theft cases?"
"Describe FIR-102"
"Show criminal network for Ramesh"

11:11 PM



Prototype Performance report/Benchmarking

Metric	Measured Score / Value	Industry Standard	Target Goal
End-to-End Latency (Query Response Time)	~1.85 seconds	3.5 – 5.0 seconds	< 2.0 seconds
Text-to-SQL Execution Accuracy	91.4%	60% – 76% (BIRD/Spider)	> 90%
RAG Groundedness (Anti-Hallucination)	98.2%	85% – 90%	> 95%
Retrieval Precision @ K=5	0.89	0.70 – 0.80	> 0.85
RBAC Security Filtering Precision	100% (Zero Leakage)	100%	100%



Provide links to your:

1. **GitHub Public Repository**-<https://github.com/Indusree21/ksp-ai-conversational-assistant.git>
2. **Demo Video Link (3 Minutes)**-
https://drive.google.com/file/d/1RoB0kQIDY7074D-sbMYbXT_Esre_tLwB/view?usp=sharing
3. **Deployed Link**-<https://ksp-ai-conversational-assistant.onrender.com>



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THANK YOU